Databases
Relational algebra

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Relational model: algebra

Relational algebra

- *fundamental* query language
- based on *set theory*
- *yardstick*: relational completeness
- *procedural*: ordering of operations
- *compositional*: a query may be composed from subqueries
- *concise*: only five basic operators
- *practical use*: query optimization

Relational algebra: selection

Unary operator: **selection** σ_p p is selection predicate

Actor		
personid	name	birth_year
nm0000204	Natalie Portman	1982
nm0000288	Christian Bale	1974
nm0000358	Daniel Day-Lewis	1957
nm0000201	Michelle Pfeiffer	1958

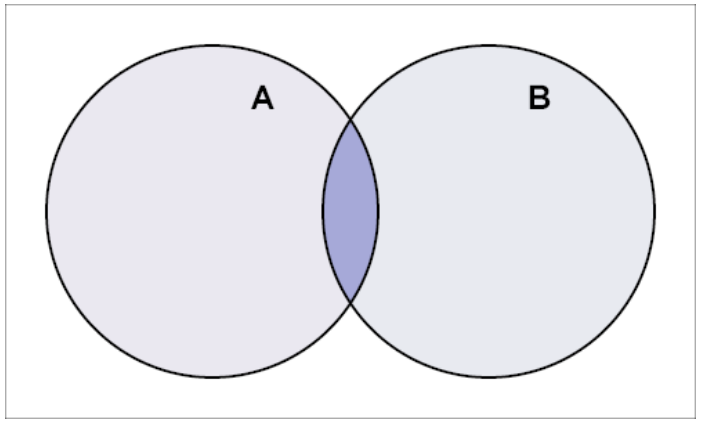
$\sigma_{birth_year<1960}$(Actor)		
personid	name	birth_year
nm0000358	Daniel Day-Lewis	1957
nm0000201	Michelle Pfeiffer	1958

Relational algebra: selection

Unary operator: **selection** with complex predicates

$$\sigma_{(birth_year<1990)\wedge(gender='female')}(\texttt{Actor})$$

Logical *and* corresponds to set intersection

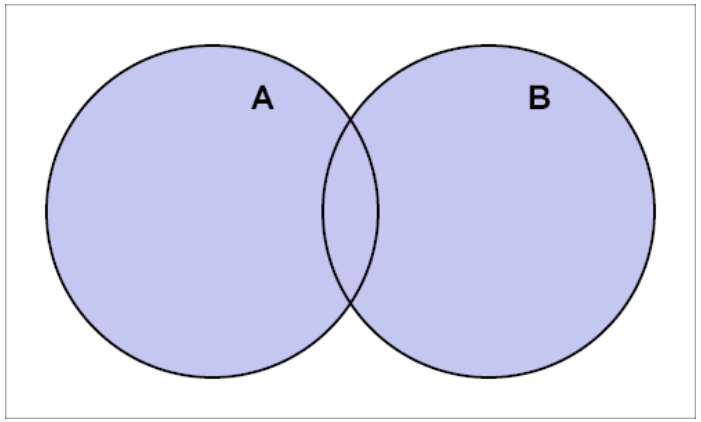


Relational algebra: selection

Unary operator: **selection** with complex predicates

$$\sigma_{(country='Netherlands')\vee(country='Belgium')}(\texttt{Actor})$$

Logical *or* corresponds to set union



Relational algebra: projection

Unary operator: **projection** π_L L is projection list

Movie			
movid	title	year	rating
tt0049366	Invasion of the Body Snatchers	1956	7,8
tt0086879	Amadeus	1984	8,4
tt0077745	Invasion of the Body Snatchers	1978	7,4
tt0078788	Apocalypse Now	1979	8,5

$\pi_{title,year}$(Movie)	
title	year
Invasion of the Body Snatchers	1956
Amadeus	1984
Invasion of the Body Snatchers	1978
Apocalypse Now	1979

Relational algebra: projection

Movie			
movid	title	year	rating
tt0049366	Invasion of the Body Snatchers	1956	7,8
tt0086879	Amadeus	1984	8,4
tt0077745	Invasion of the Body Snatchers	1978	7,4
tt0078788	Apocalypse Now	1979	8,5

π_{title}(Movie)
title
Invasion of the Body Snatchers
Amadeus
Apocalypse Now

Take care: in relational theory, tables are sets
SQL deals with duplicates (multisets or bags)

Relational algebra: composition

Composition of projection and selection

Actor		
personid	name	birth_year
nm0000204	Natalie Portman	1982
nm0000288	Christian Bale	1974
nm0000358	Daniel Day-Lewis	1957
nm0000201	Michelle Pfeiffer	1958

$\pi_{name,birth_year}(\sigma_{birth_year<1960}(\text{Actor}))$	
name	birth_year
Daniel Day-Lewis	1957
Michelle Pfeiffer	1958

Relational algebra: union

Binary operator: **union** $\cup$ schema compatibility

Actor		
personid	name	birth_year
nm0000531	Frances McDormand	1957
nm0000233	Quentin Tarantino	1963
nm0000358	Daniel Day-Lewis	1957

Director		
personid	name	birth_year
nm0000759	Paul Thomas Anderson	1970
nm0000941	Kathryn Bigelow	1951
nm0000233	Quentin Tarantino	1963

Relational algebra: union

Binary operator: **union** $\cup$ schema compatibility

Actor $\cup$ Director		
personid	name	birth_year
nm0000941	Kathryn Bigelow	1951
nm0000531	Frances McDormand	1957
nm0000358	Daniel Day-Lewis	1957
nm0000759	Paul Thomas Anderson	1970
nm0000233	Quentin Tarantino	1963

Relational algebra: difference or minus

Binary operator: **difference** − schema compatibility

Actor − Director		
personid	name	birth_year
nm0000531	Frances McDormand	1957
nm0000358	Daniel Day-Lewis	1957

Relational algebra: intersection

Binary operator: **intersection** $\cap$ schema compatibility

Actor $\cap$ Director		
personid	name	birth_year
nm0000233	Quentin Tarantino	1963

Relational algebra: Cartesian product

Binary operator: **Cartesian product** $\times$

R	
A	B
a	11
b	43

S	
C	D
b	25
c	41
b	21

R $\times$ S			
A	B	C	D
a	11	b	25
a	11	c	41
a	11	b	21
b	43	b	25
b	43	c	41
b	43	b	21

Relational algebra: theta-join

Binary operator: **theta-join** $\bowtie_\theta$

θ is matching condition

R	
A	B
a	11
b	43
c	37

S	
C	D
b	55
c	31
b	21

R $\bowtie_\theta$ S			
A	B	C	D
b	43	b	21
c	37	c	31

$$\theta : (R.A = S.C) \wedge (R.B > S.D)$$

Relational algebra: natural join

Binary operator: **natural join** $\bowtie$ default matching condition

R	
A	B
a	11
b	43
c	37

S	
A	D
b	55
c	31
b	21
d	17

R $\bowtie$ S		
A	B	D
b	43	55
b	43	21
c	37	31

Examples

Library schema: [1]

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q1: Give the names of the readers who borrowed at least one book of Dickens

[1] For simplicity, we assume that every title has only one copy in our library and we are dealing with loan history only

Examples

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q1: Give the names of the readers who borrowed at least one book of Dickens

$$\pi_{name}(Reader \bowtie (Loan \bowtie (\sigma_{author=\text{"}Dickens\text{"}} Book)))$$

Equivalence of expressions

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q1: Give the names of the readers who borrowed at least one book of Dickens

$$\pi_{name}(Reader \bowtie (Loan \bowtie (\sigma_{author=\text{"}Dickens\text{"}} Book)))$$

... but what about this one?

$$\pi_{name}(\sigma_{author=\text{"}Dickens\text{"}}(Reader \bowtie Loan \bowtie Book))$$

Examples

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q2: Give the names of the readers who never borrowed a book of Dickens

Examples

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q2: Give the names of the readers who never borrowed a book of Dickens

$$\pi_{name}(Reader \bowtie (Loan \bowtie (\sigma_{author \neq \text{"}Dickens\text{"}} Book)))$$

Note that this attempt fails. What does this expression mean?

Examples

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q2: Give the names of the readers who never borrowed a book of Dickens

First step: the completely incorrect answer

$$\pi_{rid}(Loan \bowtie (\sigma_{author=\text{"}Dickens\text{"}} Book))$$

Examples

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q2: Give the names of the readers who never borrowed a book of Dickens

Second step: take the complement of the first step

$$\pi_{rid}(Reader) - \pi_{rid}(Loan \bowtie (\sigma_{author=\text{"}Dickens\text{"}} Book))$$

To project on the names, a final join with Reader is required

Examples

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q3: Give the names of the readers who borrowed only Dickens-books

Examples

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q3: Give the names of the readers who borrowed only Dickens-books

The answer only requires a minor modification of Q2

$$\pi_{rid}(Reader) - \pi_{rid}(Loan \bowtie (\sigma_{author \neq \text{"}Dickens\text{"}} Book))$$

Examples

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q4: Give the names of the readers who borrowed all Dickens-books

???

Relational algebra: division

Binary operator: **division** $\div$

T	
A	B
a	1
a	3
a	4
b	2
b	4
c	1
c	3
d	1
d	3
d	4
d	7

U
B
1
3
4

$T \div U$
A
a
d

Relational algebra: division

Binary operator: **division** $\div$

T		
A	B	C
a	b	1
a	b	3
a	b	4
c	d	1
c	d	3
c	e	4
f	d	7
f	d	1
g	d	4
f	d	3

U
C
1
3
4

$T \div U$	
A	B
a	b

Relational algebra: division

Binary operator: **division** $\div$

- The schema of the right operand must be a strict subset of the schema of the left operand
- The schema of the resulting relation consists of the attributes of the left operand that do not occur in the schema of the right operand
- Each tuple in the resulting relation matches with all values of the right operand in the left operand of the division
- The division is not a fundamental operator; it can be expressed using projection, minus and cartesian product

Examples

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q4: Give the names of the readers who borrowed all Dickens-books

Relational model: algebra

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q4: Give the names of the readers who borrowed all Dickens-books

$$\ldots \div \pi_{bid}(\sigma_{author=\text{"}Dickens\text{"}} Book)$$

Relational model: algebra

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q4: Give the names of the readers who borrowed all Dickens-books

$$\pi_{rid,bid}(Loan) \div \pi_{bid}(\sigma_{author=\text{"}Dickens\text{"}} Book)$$

Examples

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q4: Give the names of the readers who borrowed all Dickens-books

$$Loan \div \pi_{bid}(\sigma_{author=\text{"}Dickens\text{"}} Book)$$

Why does this attempt fail?
And what is the meaning of this expression?

Relational model: algebra

Unary operators: **assignment & renaming**

$$T := < alg_expr >$$

$$T[A_1, ..., A_n] := < alg_expr >$$

Relational model: algebra

Unary operators: **renaming on the fly**

$$\rho(T)(< alg_expr >)$$

$$\rho(T, A_1, ..., A_n)(< alg_expr >)$$

Relational model: algebra

assignment & renaming examples:

$$Oldmovies1 := \pi_{movid,title}\ (\sigma_{year<1930}(Movie))$$

$$Oldmovies2[omid, omtitle] := \pi_{movid,title}\ (\sigma_{year<1930}(Movie))$$

on the fly renaming within an expression:

$$\ldots \bowtie \rho(Oldmovies, omid, omtitle)(\pi_{movid,title}\ (\sigma_{year<1930}(Movie)))$$

Relational model: algebra

Library schema:

```
Book (bid, title, author)
Reader (rid, name, address, city)
Loan (bid, rid, ldate, rdate)
```

Q5: Give the names of the readers who borrowed at least two different Dickens-books

Relational model: algebra

MonetDB: DBMS using MAL, a dialect of relational algebra

- developed at CWI, Amsterdam
- main-memory approach
- platform for analytical databases
- outperforms several commercial systems
- MAL is intermediate language for query processing
- SQL queries are translated to MAL and optimized
- *Pathfinder:* XQUERY queries are translated to MAL and optimized

Relational model: overview of algebra

Overview unary operators

- selection $\sigma_p(R)$ — p is selection predicate
- projection $\pi_L(R)$ — L is projection list
- renaming $\rho(R)$ — or using assignment

Overview binary operators

- union $R \cup S$ — schema compatibility
- difference $R - S$ — schema compatibility
- intersection $R \cap S$ — schema compatibility
- cartesian product $R \times S$
- theta-join $R \bowtie_\theta S$ — θ is matching condition
- natural join $R \bowtie S$
- division $R \div S$ — schema requirements